

Qatar-9b

R-2375

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-9_250409 · created 2026-08-20T00:43:32+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460774.677 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.134 +/- 0.055
Transit depth (Rp/Rs)^2	1.81 +/- 1.49 [%]
Orbital Inclination (inc)	84.24 +/- 1.67
Airmass coefficient 1 (a1)	2.64 +/- 0.16
Airmass coefficient 2 (a2)	-0.85 +/- 0.63
Scatter in the residuals of the lightcurve fit is	6.29 %
Best Comparison Star	#3 - [347, 207]
Optimal Method	PSF photometry
Transit Duration (day)	0.055 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.134 ± 0.055 vs 0.1489 (deviation 0.27σ)
Duration fit vs published	0.055 d vs 0.0779 d (deviation 1.43σ)
Mid-time O–C	26.9 min
Depth SNR	2.4
Residual scatter	6.29 %

Light curve

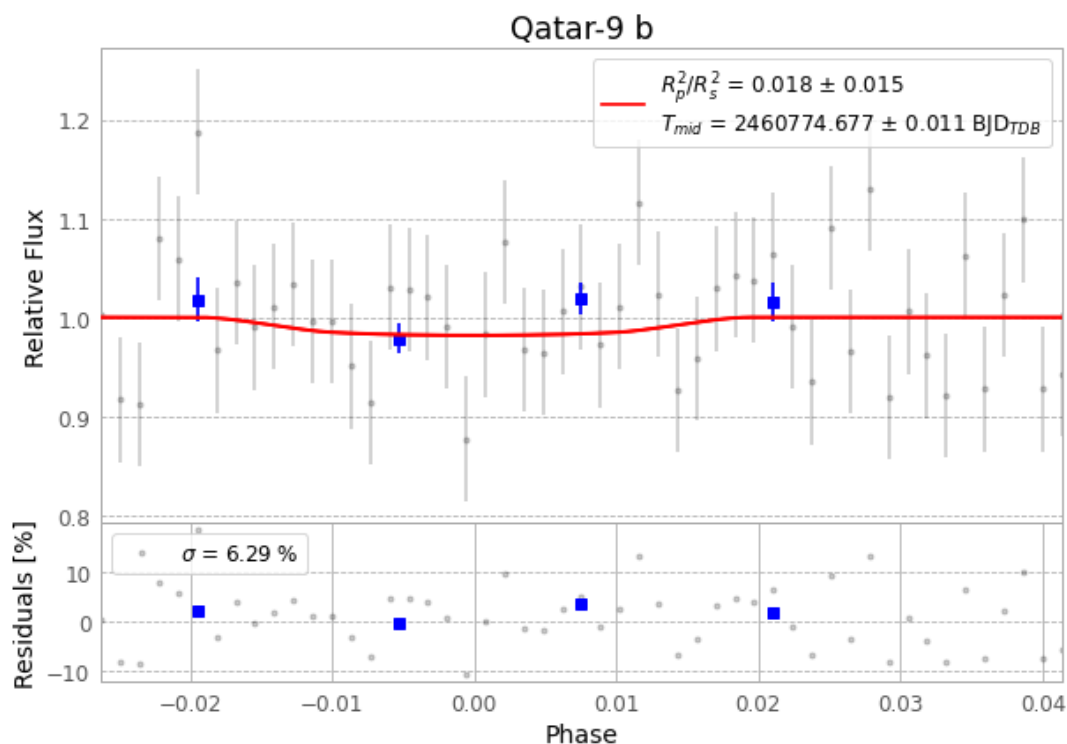


Figure 1 — FinalLightCurve_Qatar-9 b_2025-04-08.png (SHA-256 235dcf21c60d8a09...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [391, 239], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b3072fdf99098797...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

51 FITS frames (33 MB), Qatar-9250409031215.FITS → Qatar-9250409054216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-250409.log (d85a3e1f3da5...), FinalLightCurve_Qatar-9 b_2025-04-08.png (235dcf21c60d...), FinalLightCurve_Qatar-9 b_2025-04-08.csv (3e0ec3d0b85c...)

Compute resources

Wall time 97.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 00:43Z.